In this paper, we observe that typical manifolds.

We promise that: equivalent relation on X generated by: condition means the smallest equivalent relation on X satisfying the condition.

Rigorously,
$$\sim := \bigcap_{\substack{R_1 \times R_2 \subseteq X \times X \text{ is eq.rel} \\ R_1 \times R_2 \text{ satisfies the condition}}} R_1 \times R_2$$

And, $I = [0,1]$, the subspace of the euclidean space $\mathbb{R}$.

$$\begin{tikzcd}
X \arrow[d, "\pi"'] \arrow[dr, "f"] & \\
X/\sim \arrow[r, "\tilde{f}"'] & Y
\end{tikzcd}$$

「Cylinder

「Def.
Define an equivalent relation $\sim$ on $I \times I$ generated by : $\forall y \in I,\ (0,y) \sim (1,y)$
The quotient space $I^2/\sim$ is called the Cylinder.

「Prop. The Cylinder $I^2/\sim$ is Homeomorphic to $S^1 \times I \subseteq \mathbb{C} \times \mathbb{R}$.
Proof. Define a map

$$f : I^2 \to S^1 \times I : (x,y) \mapsto (e^{2\pi i \cdot x}, y)$$

$$\begin{tikzcd} I^2 \arrow[d, "\pi"'] \arrow[dr, "f"] & \\ I^2/\sim \arrow[r, "\tilde{f}"'] & S^1 \times I \end{tikzcd}$$

Then, f is Continuous onto closed map, because

1. Continuous
$(x,y) \mapsto e^{2\pi i \cdot x}$ and $(x,y) \mapsto y$ are Continuous, so f is also Continuous.

2. Onto
Given $(e^{i\theta}, b) \in S^1 \times I$ where $\theta \in [0, 2\pi]$, take $x = \frac{\theta}{2\pi}$ and $y = b$, then clear.

3. Closed
Given closed set $C \subseteq I^2$, C is Compact since I^2 is compact.
So, the Continuous image $f[C]$ is also Compact in $S^1 \times I$,
therefore $f[C]$ is closed, by $S^1 \times I$ is Hausdorff.

Moreover, $f(x_1, y_1) = f(x_2, y_2) \iff e^{2\pi i x_1} = e^{2\pi i x_2}$ and $y_1 = y_2$
$(\iff e^{2\pi i (x_1 - x_2)} = 1)$

$$\iff \left(x_1 = x_2 \text{ or } \{x_1, x_2\} = \{0, 1\} \right) \text{ and } y_1 = y_2$$

$$\iff (x_1, y_1) \sim (x_2, y_2)$$

Therefore, f induces a Homeomorphism

$$\tilde{f} : I^2/\sim \to S^1 \times I : [(x,y)] \mapsto f(x,y).$$

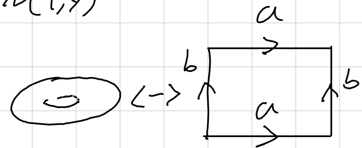
It is injective and Well-defined by $(x_1, y_1) \sim (x_2, y_2) \iff f(x_1, y_1) = f(x_2, y_2)$
And surjective is given by the f is surjective.
Finally, the Continuity is given by : $\tilde{f} \circ \pi = f$ is continuous, and,
given closed set $C \subseteq I^2/\sim$, $\pi^{-1}[C]$ is closed, so the ontoness of π gives
$\tilde{f}[C] = \tilde{f}[\pi[\pi^{-1}[C]]] = f[\pi^{-1}[C]]$ is closed, since f is closed map. ☒

Torus

Def.

Define an equivalent relation $\sim$ on $I \times I$ generated by:

$$\begin{cases} \forall x, y \in I, \ (x,0) \sim (x,1) \text{ and } (0,y) \sim (1,y) \\ (0,0) \sim (1,0) \sim (0,1) \sim (1,1) \end{cases}$$



The quotient space $I^2/\sim$ is called a Torus.

$\langle a, b \mid aba^{-1}b^{-1} = e \rangle$
$\cong \mathbb{Z} \times \mathbb{Z}$

$I^2/\sim \; \cong S^1 \times S^1$ gives
$\pi_1(I^2/\sim) \cong \pi_1(S^1 \times S^1)$
$\cong \pi_1(S^1) \times \pi_1(S^1)$

Prop. The Torus $I^2/\sim$ is Homeomorphic to $S^1 \times S^1 \subseteq \mathbb{C}$.

Proof. Define a map

$$f : I^2 \to S^2 : (x, y) \mapsto (e^{2\pi i \cdot x}, e^{2\pi i \cdot y})$$

Then, f is continuous onto closed map,

Projective Space.

Def. Define an equivalent relation on $\mathbb{R}^{n+1}\setminus\{0\}$ generated by:

$$x \sim y \iff \exists t \in \mathbb{R} \text{ such that } tx = y.$$

The quotient space $\mathbb{R}P^n \overset{\text{def}}{=} (\mathbb{R}^{n+1}\setminus\{0\})/\sim$ is called a Real Projective Space.

Prop. $\mathbb{R}P^1$ is Homeomorphic to the Circle S^1.

Proof. Define a map

$$f : \mathbb{R}^2\setminus\{(0,0)\} \to S^1 : (x,y) \mapsto \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}}\right)$$

Then, it is continuous clearly, and onto: given $(\cos\theta, \sin\theta) \in S^1$, take

$$x = \cos\theta \quad \text{and} \quad y = \sin\theta.$$

Moreover,

$$(x_1,y_1) \sim (x_2,y_2) \iff \exists t \in \mathbb{R} \text{ such that } (x_1,y_1) = (tx_2, ty_2)$$

$$\iff f(x_1,y_1) = f(x_2,y_2)$$

Therefore, f induces the Homeomorphism

$$\bar{f} : \mathbb{R}P^1 \to S^1 : [(x,y)] \mapsto f(x,y).$$

So proved. $\square$

Γ Klein bottle.

Möbius band.